

IC SPICY

IC SPICY Agent Swarm Worker Setup

Step-by-step guide to run the off-chain agent worker on Mac or Chromebook.

August 2026 • Mobile-friendly operator guide

What You Are Building

The Agent Swarm has two parts: agent_hub (on-chain brain on ICP) and a Node.js worker (off-chain hands). You control both. The worker uses a BIP39 seed phrase to derive a principal ? register that principal in Admin ? Agent Swarm ? Worker principals.

The hub stores jobs, drafts, approvals, subscribers, secrets, and audit logs. The worker polls for jobs, reads backend data (recipes, orders, weather), calls your LLM API, and submits drafts for your approval before anything is sent.

Never store API keys in git. Only the seed phrase lives on the worker machine (in .env or a file with chmod 600). Resend and LLM keys are set in Admin ? Agent Swarm ? Secrets.

Prerequisites

Node.js 20+ (nvm recommended on Mac; Crostini Linux on Chromebook).

- dfx 0.29+ and mops for canister deploys (already in this repo).

Resend account (resend.com) with a verified sending domain DNS records.

LLM API key: OpenAI or Anthropic (SpicyAI on-chain fallback is rate-limited).

Admin access to icspicy.app Admin ? Agent Swarm tab.

~3T cycles for agent_hub canister on mainnet (one-time create).

Step 1 ? Generate Worker Identity

- cd worker && cp .env.example .env

Generate a new 12-word BIP39 mnemonic (write it on paper; never commit it): node -e "console.log(require('bip39').generateMnemonic())"

Set WORKER_MNEMONIC="word1 word2 ..." in .env OR save words to ~/.ic-spicy/worker-mnemonic.txt and set WORKER_MNEMONIC_FILE.

- npm install && npm run show-principal ? copy the printed principal.

In Admin ? Agent Swarm ? Worker principals ? Register that principal. Also register it on backend via backend addAgentPrincipal (same Admin flow uses hub; backend agent role enables fleet/orders reads).

Step 2 ? Configure Secrets (Admin UI)

Admin ? Agent Swarm ? Secrets. Set: resend_api_key, resend_from_email (e.g. newsletter@yourdomain.com), llm_api_key, llm_provider (openai or anthropic), llm_model (e.g. gpt-4o-mini), admin_alert_email (optional), newsletter_reply_to (optional).

Set AGENT_HUB_CANISTER_ID and BACKEND_CANISTER_ID in worker/.env after deploy (see canister_ids.json).

Set PUBLIC_SITE_URL to your frontend URL for newsletter confirm links (mainnet: <https://7rukvhqaaa-aaaao-ba6ma-cai.icp0.io>).

Step 3 ? M1 MacBook Air Setup

Install nvm: curl -o- <https://raw.githubusercontent.com/nvm-sh/nvm/v0.40.1/install.sh> | bash

- nvm install 20 && nvm use 20

Clone or open this repo. cd worker && npm install

Test one poll cycle: npm run once (requires local or mainnet canisters deployed and worker principal registered).

Always-on: edit worker/deploy/com.icspicy.agent-worker.plist ? replace CHANGE_ME paths with your repo path. mkdir worker/logs

- launchctl load
~/Library/LaunchAgents/com.icspicy.agent-worker.plist (copy plist there first).

Monitor: tail -f worker/logs/stdout.log

Step 4 ? Chromebook (Crostini Linux)

Settings ? Developers ? Linux ? Turn on. Open Terminal.

- sudo apt update && sudo apt install -y curl git build-essential

Install nvm and Node 20 (same commands as Mac).

Clone repo into Linux files (~/.IC-SPICY-MAIN). cd worker && npm install

Prevent sleep: Settings ? Device ? Power ? Sleep when inactive ? Never (while plugged in).

Copy worker/deploy/ic-spicy-agent-worker.service ? replace CHANGE_ME paths and User.

- sudo cp ic-spicy-agent-worker.service /etc/systemd/system/ && sudo systemctl daemon-reload
- sudo systemctl enable --now ic-spicy-agent-worker.service

Check status: sudo systemctl status ic-spicy-agent-worker

Step 5 ? Resend DNS Verification

In Resend dashboard ? Domains ? Add domain (your sending domain).

Add the SPF, DKIM, and optional DMARC DNS records Resend provides.

Wait for verification (usually minutes to hours).

Use the verified address as `resend_from_email` in Admin secrets.

Test: subscribe your email via site footer ? confirm link
? approve newsletter draft in Admin ? worker sends on next poll.

Step 6 ? Daily Workflow

Orchestrator enqueues jobs on a timer inside `agent_hub`.

Worker claims jobs every `POLL_INTERVAL_SEC` (default 60s).

Social agents submit drafts only ? review in Admin ? Agent Swarm ? Approval Queue. Approve, edit, or reject.

Newsletter agent creates HTML draft weekly. After approval, worker sends via Resend with idempotency keys (no double-send on retry).

Ops agents (fleet/cycles, weather, orders/claims, analytics) post internal digests to the same queue.

Compliance reviewer runs rule checks + optional LLM review before drafts enter the queue.

Step 7 ? Deploy Canisters

Local first: `dfx start --background --clean && dfx deploy --network local backend agent_hub`

- `dfx generate backend && pnpm bindgen` from repo root.

Frontend: `cd src/frontend && pnpm build`. Set `CANISTER_ID_AGENT_HUB` in env for Vite.

Mainnet (requires explicit confirmation): `dfx deploy --network ic agent_hub (~3T cycles)`, then `dfx deploy --network ic backend`, then `node scripts/upload-frontend.mjs --network ic --canister 7rukvhqaaa-aaaao-ba6ma-cai`

Post-deploy: register worker principal, set secrets, subscribe test email, run `npm run` once on worker.

Troubleshooting

agent only / admin or agent only: worker principal not registered in hub or backend.

Newsletter service not configured: `CANISTER_ID_AGENT_HUB` missing in frontend build env.

Resend 403: domain not verified or wrong from address.

LLM errors: check llm_api_key and provider/model in Admin secrets.

Revoke compromised worker: Admin ? remove worker principal immediately; rotate mnemonic and re-register.

Security Checklist

Seed phrase: paper backup only; chmod 600 on .env and mnemonic file.

Worker is agent role, never full admin.

All public sends require human approval in Admin queue (start here; auto-post later per platform).

Audit log in Admin ? Agent Swarm ? Audit for every draft and secret change.

Do not log customer email content in worker stdout in production.